

Mapping Player Mobility and Club Connectivity in World Football Using Network Science

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1. Abstract

The main purpose of this project is to analyze temporal and structural patterns from professional football using network science. We will construct and investigate two networks derived from a large dataset of players, clubs and transfer over a span of almost three decades. A player-club bi-partite network will be studied which captures the static relationships between players and their clubs, while a directed club-club transfer network represents the movement of players across teams. The transfer network is additionally studied with the target of uncovering its evolution in time. Using metrics such as degree distribution or detection of the communities will help us discovering transfer patterns and to identify most influential clubs across in Europe. This study demonstrates how network analysis can effectively model and interpret complex systems in sports, offering insights into transfer market dynamics, league structure, and player mobility.

2. Introduction

Football transfers create a large network where clubs can transfer players across different leagues and countries. This happens twice per year, a transfer period(usually called "mercato period" by football fans) which is in the summer and the other one in the winter. These complex links between players, clubs, amount of money paid for a player and many other arguments made us understood that network science can offer an useful way of study this type of data, because it focuses on how different entities(clubs, players) are connected and how these connections change over time.

In this project, we used real data [1] from only world football to build two main networks. The first one is a player-club bipartite network, which shows how players are currently distributed across clubs. The second one is a directed club-club transfer network, where each edge represents a player moving from one club to another, and the edge weight corresponds to the number of transfers that occurred between that pair of clubs.

An additional information that our dataset contains is transfer's date which can be useful in order to discover and discuss how the transfer networks evolves over periods of time(ex: 5 years each one). This helps understand better whether certain clubs became more central, and how movement between leagues has changed.

Overall, the goal of the project is to apply network science tools(such as degree analysis temporal comparisons), because many patterns can be hidden behind simple transfers. The results show how networks help understand trends and changes in a system that is constantly changing.

3. Data Preprocessing

First of all, like in any other project which is built relied on data, preprocessing step is a need. The initial dataset downloaded from Kaagle [1] contained many useless CSVs for the main purpose of the study and starting to work on networks creation and discovering metrics would have been not possible without data preprocessing step.

Seven files were removed (appearances.csv, clubgames.csv, competitions.csv, gameevents.csv, gamelineups.csv, games.csv, playervaluation.csv) to ensure that all that has been kept is enough for the project. Furthermore, the remaining files(clubs.csv, transfers.csv, players.csv) included many attributes that were not so relevant for our analysis, so the first step was to reduce each table to only columns needed to build the player-club and club-club networks.

```
player_keep = [
    "player_id",
    "name",
    "country_of_citizenship",
    "position",
    "current_club_id",
    "market_value_in_eur",
]
```

Figure 1: Attributes kept for players

The above attributes that can be seen in Figure 1 are enough to construct the bipartite network and allow us to analyze club composition, nationality patterns and player distribution. Afterwards, there were also removed duplicate player entries and player_id was converted into integer on purpose. At the same time, player's market value was converted into numeric. Players without a current club were removed from the dataset since they do not contribute to either of the networks.

For clubs, it was selected a small subset of attributes necessary for identifying clubs and linking them to leagues such as

club ID, name and domestic league where it is currently registered. In the end, transfer CSV which is the spine of second network has been cleaned in the same way. First, there were kept only the necessarily attributes (player ID, from what to what club the transfer has been made and transfer date). However, several modification were applied such as parsing data, removing rows with invalid or missing information or filtering the clubs where so that only moves between clubs present in clubs.csv were kept.

4. Network Construction and Analysis

4.1. Bipartite Club-Player Network

The bipartite club-player network was built using players and clubs datasets, which contain the current club information for each players and the corresponding club identifiers, names, and competitions levels. As it was already mentioned above, only players whose club_current_id appears in the cleaned club list were recorded, thus we avoided invalid associations. An important fact is that each player appears only with the club they most recently played for in their career. By this restriction each player to their last known club, the network becomes a clean representation of the current period and also allow us to study how many players each club has and financial structure of the club (via market value weights). We use only the most recent club for each player to obtain a clean snapshot of the current football structure, while keeping career movements and historical dynamics strictly within the transfer network.

The network has one layer which represents players and the other layer which represents clubs, edges connect a player to the club who currently belong to. Basically, this yields a classical player-club affiliation network. Since each player belongs to only one club, the resulting bi-partite network is star-like: each club connects to multiple players and each player connects to exactly one club.

Down below, it can be seen a smaller dimensions representation for my graph (only top 10 clubs were used in terms of number of players registered).

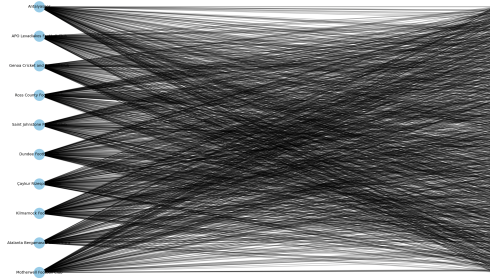


Figure 2: The shape of the first graph (smaller version, using 10 clubs)

In Table 1, it summarizes the size of the player-club bipartite network. Each player is linked only to their most recent club, resulting in 32,601 player-club edges across 32,601 players and 439 clubs.

Property	Value
Total nodes in bipartite network	33,040
Number of player nodes	32,601
Number of club nodes	439
Total edges (player-club connections)	32,601

Table 1: Basic size statistics of the player-club bipartite network

4.2. Directed Club-Club Transfer Network

For the second network, we used the transfer_clean.csv and clubs_clean.csv. The transfer contains for each move, the player identifier, the club_id from where he came and the destination club which was recorded as to_club_id and the year of the transfer (this information is useful, subsequently for comparisons over periods of time).

Nodes are represented by clubs and a directed edge $u \rightarrow v$ (where u and v are considered clubs) is linked if at least one player moved from club u to club v in the selected time period. Moreover, the weights are computed as number of transfer between those clubs, obtained by using the transfer records (from_club_id, to_club_id) and counting them. In the results, apart of the original network which displays results over 23 years, it will be discussed the fact that we computed networks for each period with a span of 5 years to observe how transfer market evolved and computing metrics for each period (starting with 2002 until 2026). This allows us to examine how the transfer market has grown over time, whether central clubs become more central in later periods and what players had most transfers over the years.

The evolution of the World football transfer ecosystem can be better understood by examining these temporal slices, which also reveal long-term patterns that are not apparent in the aggregated network.

5. Results and Discussion

5.1. Bipartite Club-Player Network

First of all, squad-size distribution is highly uneven across clubs. The maximum degree is 42 players, while the mean squad size is 32, indicating a substantial right-skew in the data: a small number of clubs field significantly larger squads than the typical club in the dataset (that can be noticed in Table 2). The top clubs by squad size all originate from a limited set of leagues (most notably the Scottish Premiership) suggesting that certain competitions retain comparatively larger rosters, possibly due to differences in registration rules, youth-team integration (all these information can be seen in Table 2).

Club	League	Players
Kilmarnock FC	SC1	42
Motherwell FC	SC1	41
Ross County FC	SC1	41
Dundee FC	SC1	39
Çaykur Rizespor	TR1	38

Table 2: Top 10 clubs by squad size

Second of all, in Table 3 market value for each club which was granted by summing market value for their players(clubs strength). In the table below it could be seen that clubs like Manchester City, Real Madrid, Arsenal, Barcelona dominate the ranking, each with squad valuations above €1 billion. This imposes the next fact: large squads do not imply high squad value, and the richest clubs maintain compact but highly valuable rosters.

Club	League	Total Value (€)
Manchester City FC	GB1	1,374,875,000
Real Madrid CF	ES1	1,325,525,000
Arsenal FC	GB1	1,178,225,000
FC Barcelona	ES1	1,153,525,000
Liverpool FC	GB1	1,100,275,000
Paris Saint-Germain	FR1	1,058,800,000
Chelsea FC	GB1	1,017,575,000

Table 3: Top 7 clubs by total squad market value

Moreover, we computed the average market value per player, and here the rankings really became more interesting. Clubs like Real Madrid or Manchester City are leading with an average above 18M euro per player. This is a normal behavior, because the clubs with a less average by number of players lead the top of total squad market values and they have the most expensive players.

Club	League	Players	Avg Value (€)
Real Madrid CF	ES1	26	27,051,530
Manchester City FC	GB1	26	23,302,966
FC Barcelona	ES1	23	18,605,242
Paris Saint-Germain	FR1	24	18,255,172
Arsenal FC	GB1	25	15,709,666
Liverpool FC	GB1	26	14,289,286
FC Bayern München	L1	25	13,039,084

Table 4: Top 7 clubs by average player market value

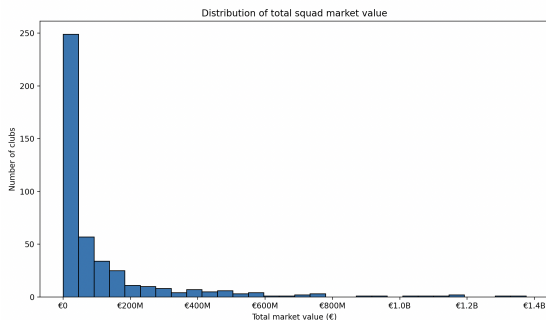


Figure 3: Distribution of total squad analysis

The distribution of total squad market value (shown in Figure 3) is extremely right-skewed, with most clubs concentrated near the lower end and a few clubs reaching values above €1 billion and that is quite normal pattern for professional football,

because there are not so many clubs with a squad total value which exceeds €1 billion. In terms of network, the weighted degree distribution is not uniform or normal, it resembles a distribution where many clubs have low total player value and it is a very small number of clubs which could be considered hubs ("high-weighted").

Because the wealthiest clubs usually attract the highest-value players, the bipartite network ends up being strongly shaped by market value. In practice, this creates positive strength assortativity, where expensive players are mostly linked to already valuable clubs. We can also see a clear degree–strength pattern: the richest clubs do not need large squads, but instead rely on a smaller set of very high-value players. This shows that the structure of the network is driven more by player value than by the number of connections.

5.2. Directed Club-Club Transfer Network

To complement the player–club analysis, we next examine the transfer market through a directed club–club network, which is more dynamic.

Over the full 2002-2026 period(we consider 2026 for the player who didn't renew their contract yet),the directed club-club transfer network contains 439 clubs and 12920 directed links, which we found it dense. On average, a club is linked with other 29 distinct clubs(mean out-degree of 29) and is involved in about 47 outgoing transfers in total(mean out-strength 47.4). That still confirms that transfer markets forms a well-connected network, where most clubs interact with a few partners. Down below, in Table 5, it can be observed all discussed above.

Property	Value
Number of clubs (nodes)	439
Number of directed transfer links (edges)	12,920
Observation window (years)	2002–2026
Mean out-degree	29.43
Mean out-strength (players sent)	47.42

Table 5: Statistics of the directed club-club transfer network

The main hubs of this network act as the main sellers or receivers on the transfer market in terms of clubs. These clubs sit near the core of the transfer graph and are important for maintaining global connectivity, since they bridge many otherwise weakly connected parts of the network. Below, it was shown which are the top 5 clubs(in and out transfers).

Club	League	Sold players
SL Benfica	PO1	103
Chelsea FC	GB1	89
Inter Milan	IT1	85
Olympiakos	GR1	85
Manchester City FC	GB1	82

Table 6: Top 5 clubs by number of distinct destinations (out-degree)

Club	League	Bought players
SL Benfica	PO1	103
Chelsea FC	GB1	95
Galatasaray	TR1	92
Inter Milan	IT1	90
Olympiakos	GR1	84

Table 7: Top 5 clubs by number of distinct sources (in-degree)

Apart from the network field, there are interesting result which are comparable with the real ones. SL Benfica is considered one of the best sellers in the world and this can be found here [2]. This means that our results are on the right track. An exciting fact is also that SL Benfica and Chelsea are on the first and second position at this top either as importer or exporter

Furthermore, we turned our network into an undirected graph so that there could be observed communities. We used modularity optimisation to discover communities, the three largest communities found in group 107, 86 and 75 clubs respectively. We could interpret these communities as "ecosystems" where players have a bigger chance to be transferred. This is an important statistic in terms of football, because in earlier times using these strong relationship between clubs considered in the same ecosystem(community), chances to get a great deal were being higher rather than trying to buy a player from out of you ecosystem.

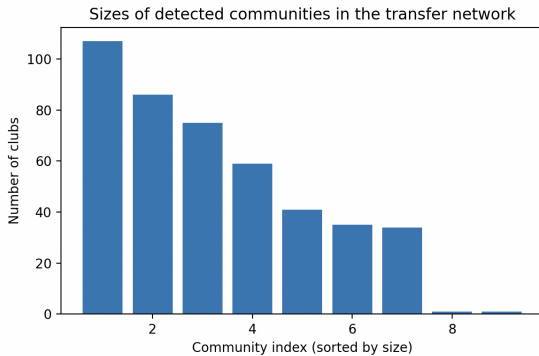


Figure 4: Distribution of total squad analysis

As illustrated above, there were found 9 communities in our network which is quite a small number, but there is something important that should be mentioned about which clubs are included in those communities. A strong regional and league-based pattern: one community mixes many Turkish, Greek and Portuguese clubs together, another contains a large number of English and Spanish teams, and a third combines German, Dutch and Danish clubs. In network science terms this means the transfer market is modular: clubs tend to trade more frequently inside regional or league based blocks than across them(this is what I discussed above about ecosystems). We associated these communities with continents, according to the results given and the ecosystem's compound, there is a real concentration among teams from the same continent.

In the next paragraph, we will summarize the evolution of

transfer market over the last 23 years using 5 years as block of time:

Window	Edges	Comment
2002–2006	36	early, very sparse market(dataset limit)
2007–2011	511	market becomes clearly connected
2012–2016	2,431	strong growth in transfer activity
2017–2021	5,783	dense, highly active network
2022–2026	6,980	further expansion in recent years

Table 8: Number of directed transfer links per 5-year window.

As is shown in the Table 8, the number of edges(which are considered number of transfers) increased period by period and this is very normal. There is a quite normal because of the limitation of this project(we do not have so many transfer data from 2002 to 2010), but starting with 2011, there are full number of transfers from [3] being the official website which FIFA (international football federation) works with. This shows a clear densification of the transfer network: over time, more pairs of clubs become connected and the market becomes structurally more integrated.

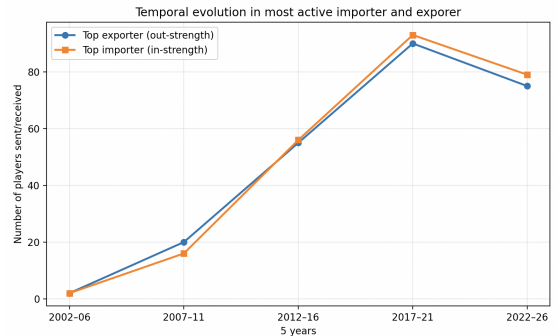


Figure 5: Top Export/Import over the years

Both the maximum in-strength and out-strength increase drastically over time(the reasons are discussed above). While the best exporters and importers may handle 70–90 moves per period in later windows, the most active teams only participate in a few transfers in early windows. This implies that the transfer network's hubs grow stronger and concentrate a growing number of player flows.

In figure 6, is is shown how the number of transfers per period evolved for one player and we can introduce a metric called the mobility of the player which is computed as a simple count in number of transfers. The player with the highest mobility over the years is Roberto Piccoli with 7 transfers(in 5 years).

Overall, the temporal analysis confirms that the club–club transfer network is not static. The number of edges grows rapidly, the maximum values for in and out increases and the identity of the most central clubs changes from period to period (from Portuguese and English clubs in the early 2000s to Italian, Greek and other clubs in later periods). The global structure becomes more clustered and this indicates a denser network.

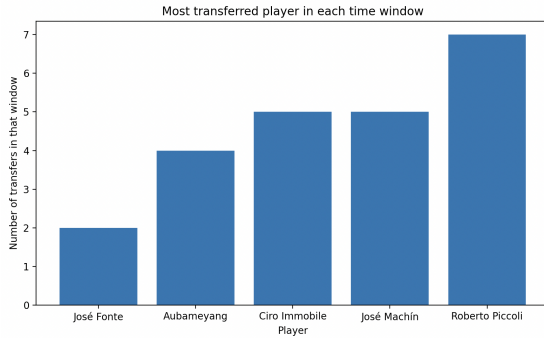


Figure 6: Number of most transfers/player over the years

5.3. Why do we not use certain metrics?

This part was not mandatory for the report but adding a short discussion about not using clustering coefficient (which was well understood from the lectures) made more clear the fact that our networks do not need the computation of clustering coefficient. However, this metric is not informative for the player–club bipartite network, because bipartite graphs cannot contain triangles by construction; thus, their clustering coefficient is always zero and does not reflect any meaningful structure. For the directed club–club transfer network, clustering is also limited, since directed transfer flows rarely form triangular patterns in practice and this are either transfers made on purpose, either pure chance. For these reasons, clustering is not used as a metric in our analysis.

6. Limitations

One issue with the dataset is that the first and second period(2002–2011) contains very few recorded transfers. This makes the resulting network extremely sparse and much less representative than the later windows, where transfer activity is documented more consistently. Even though the data is incomplete, we still include this period because it helps show how the network evolves over time and highlights the contrast between an under-recorded early market and a fully developed later one.

Another limitation comes from the fact that some movements which are between youth academy of one clubs and senior team(this can affect some of the transfers for clubs which have a numerous academy). These are not real interactions between clubs on transfer market but, they have been kept due to the limited number of cases.

The analysis also leaves out all past transfers involving retired players. Clubs that were historically very active might seem less central simply because many of their former players are no longer part of today’s dataset. This choice avoids artificially adding outdated connections, but it means that the network mostly captures the modern transfer landscape.

7. Future work

A natural extension of this study can be building a dynamic multilayer network, where each period of time lasts long for 1

year and edges evolve over time. This will allow studying how centrality changes from year to year or how shocks (Covid-19) affects transfer market.

An additional enhancement can be integrating player features such as: age, nationality or position to check if certain clubs have different patterns of transfers, but this links directly to attribute-driven network models.

At the same time, it can be added the graph neural network approach. Using this Machine Learning model, a few extra improvements such as predictions for transfers in the future will be a great idea of development.

8. Conclusion

In conclusion, in this project we mapped the structure of professional football using two main networks: a weighted bipartite club-player network and a directed club-club network. Together, these models captured two different perspectives of the football ecosystem: the static snapshot of where players currently play, and the dynamic flow of players between clubs over more than two decades(from 2002 to 2026).

The analysis of the bipartite graph was shown a highly uneven structure, where a small number of elite clubs have a high total market value despite small number of players in their squads(but this is quite normal in football).Furthermore, using these approach, there was discovered a pattern that was already know: clubs rely on large squads with modest value, while only a few teams operate as high-value hubs.

The directed transfer network added the temporal and directional dimension needed to understand how clubs actually interact and how the relationship between them has changed over the last 23 years.The transfer market became significantly denser: from only a few dozen club-to-club edges in 2002–2006 to several thousand in the most recent periods. This indicates increased mobility and globalisation in football.

Finally, this work showed how network science can model a complex sports environment, offering a structured way to explore inequality, connectivity, community structure, and temporal evolution in the global football market.

References

- [1] D. Cariboo, Player transfer values and market data, <https://www.kaggle.com/datasets/davidcariboo/player-scores>, kaggle Dataset (2025).
- [2] P. Fitzpatrick, Benfica are the greatest salesmen in football; even brighton cannot touch them, accessed: 2025-11-20 (2023).
URL <https://www.football365.com/news/benfica-greatest-salesmen-ramos-fernandez-nunez>
- [3] Tranfermarkt, accessed: 2025-11-20 (2025).
URL <https://www.transfermarkt.com/>